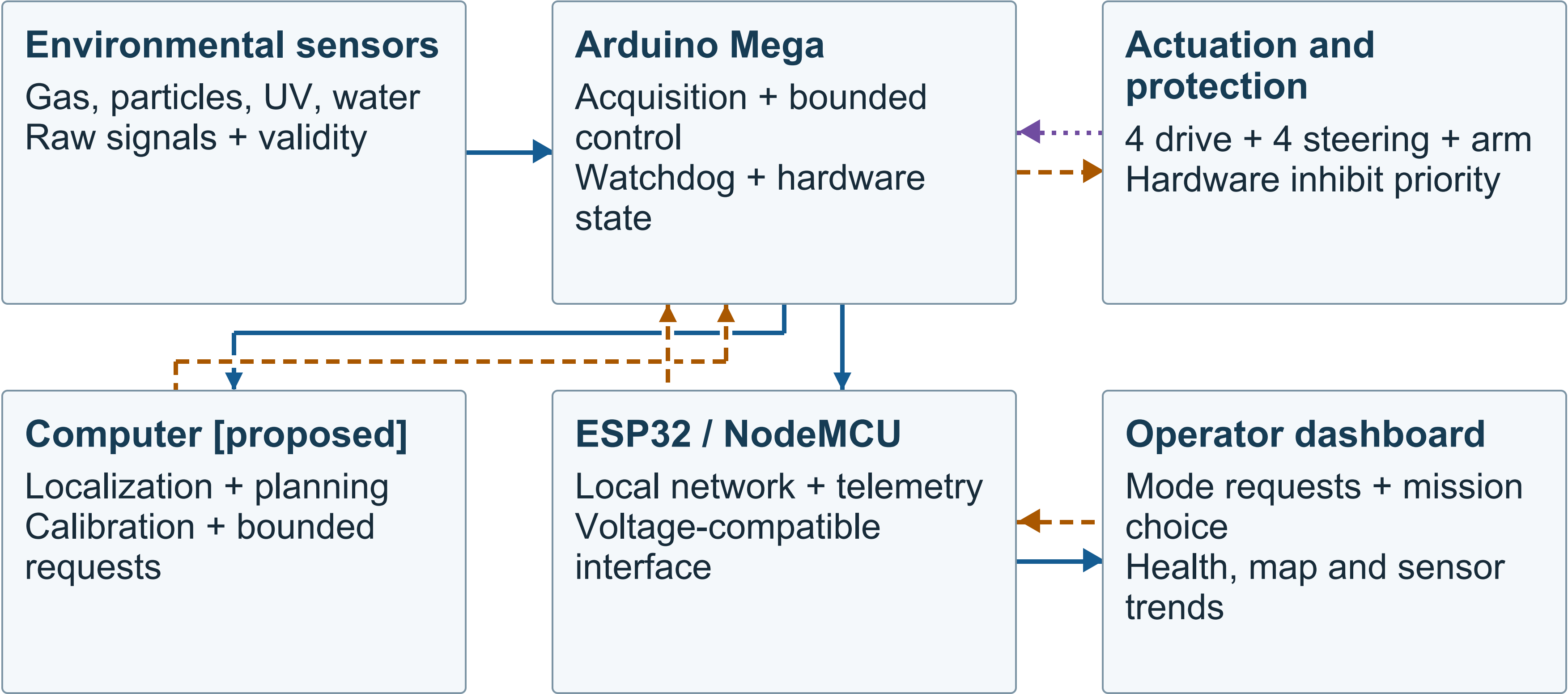


Figure 8. System control architecture

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

—→ Measurement or data record

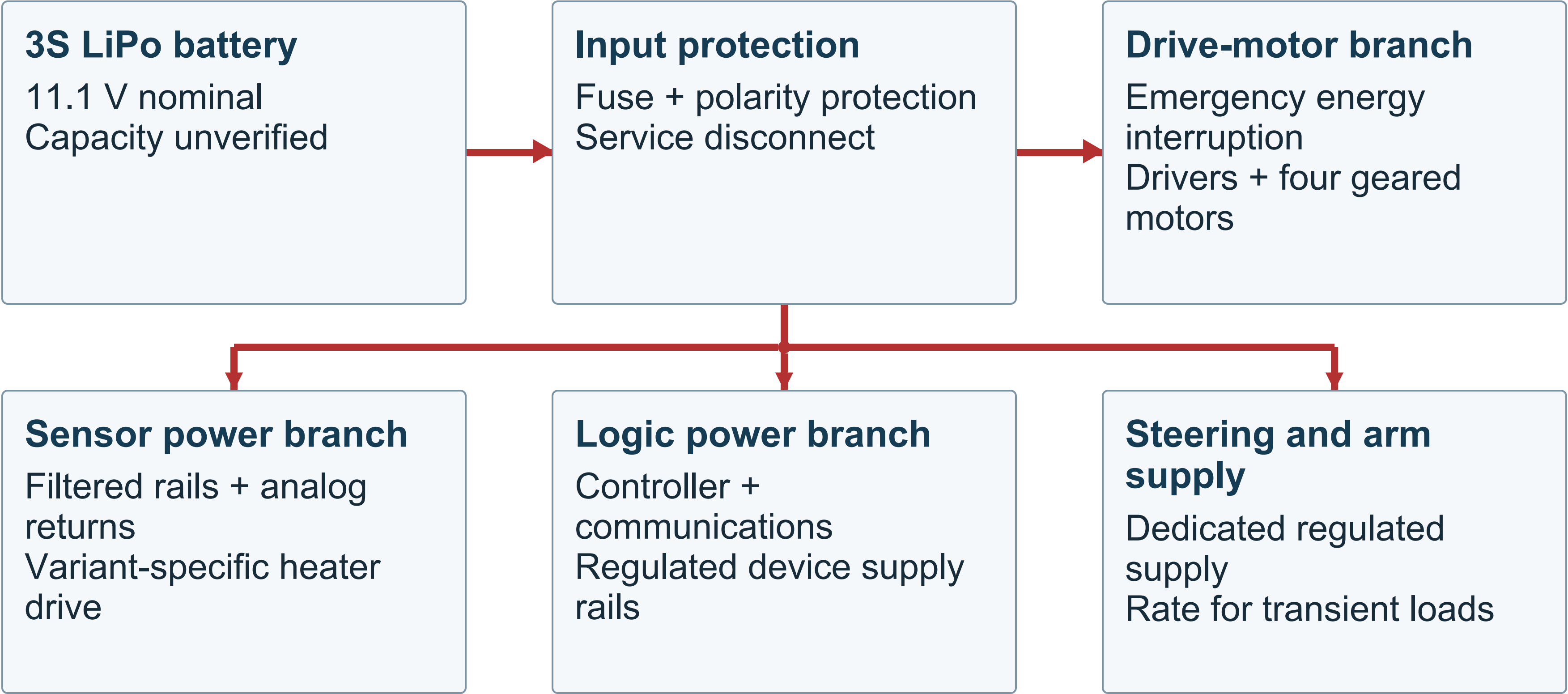
- - → Command or inhibit request

...→ Measured state returned to control

Functional links require verification. Crossings without a junction dot are not connected.

Figure 28. Protected power distribution

A3P5 NEMESIS | Proposed functional architecture



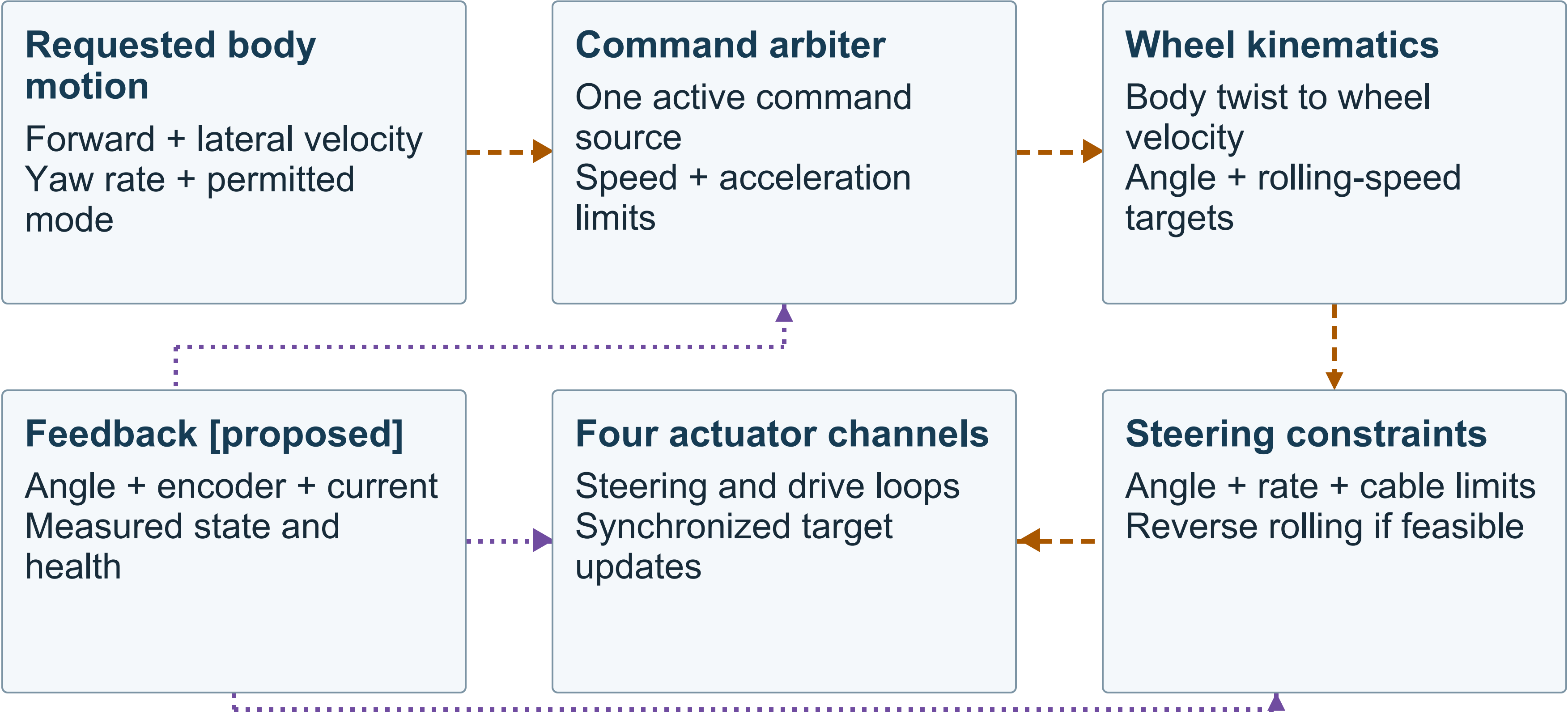
LEGEND · Arrow direction indicates the direction of the link

➡ Electrical power supply

Parallel load branches; dots mark common supply junctions. Returns and pinouts are omitted.

Figure 9. Four-wheel command allocation

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

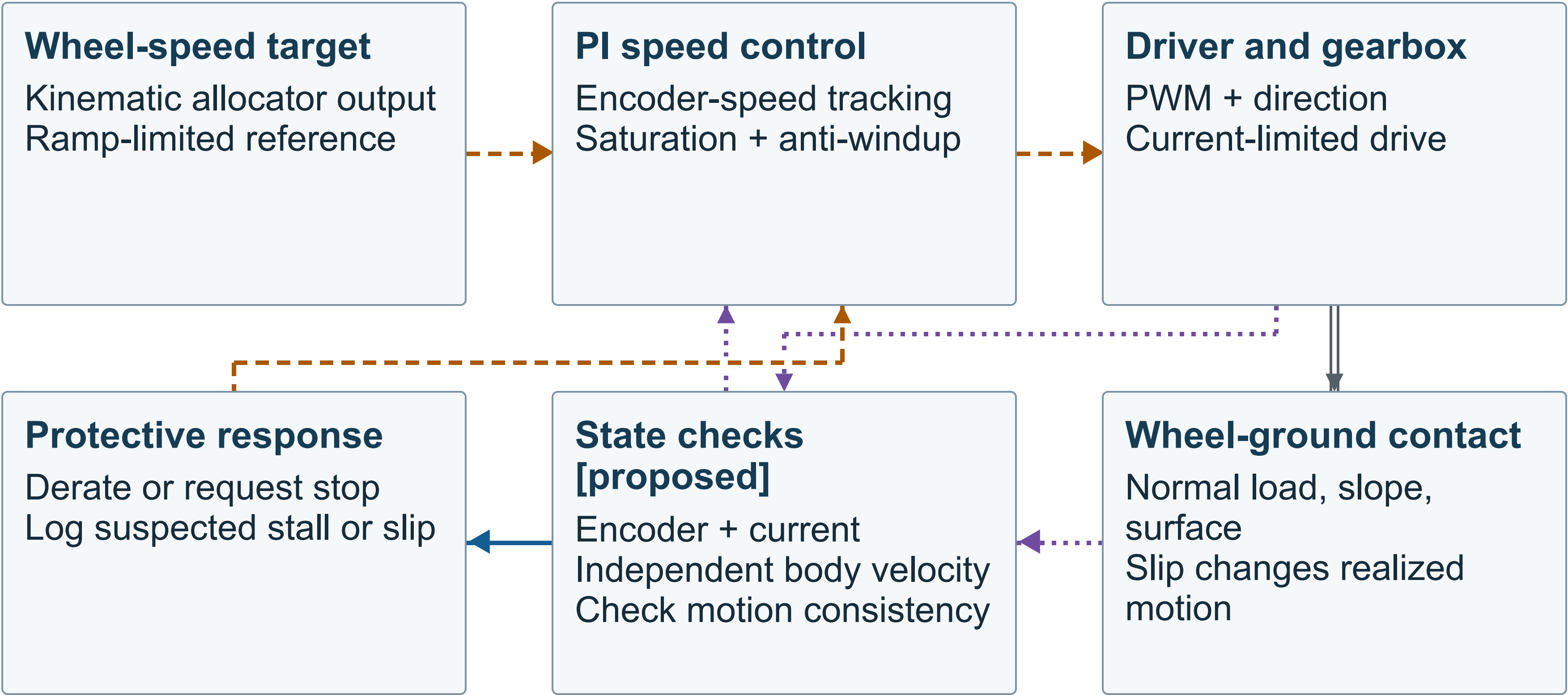
- - -> Command or inhibit request

.....> Measured state returned to control

Functional links require verification. Crossings without a junction dot are not connected.

Figure 11. Drive control and traction monitoring

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

---➔ Command or inhibit request

==➔ Physical transfer or interaction

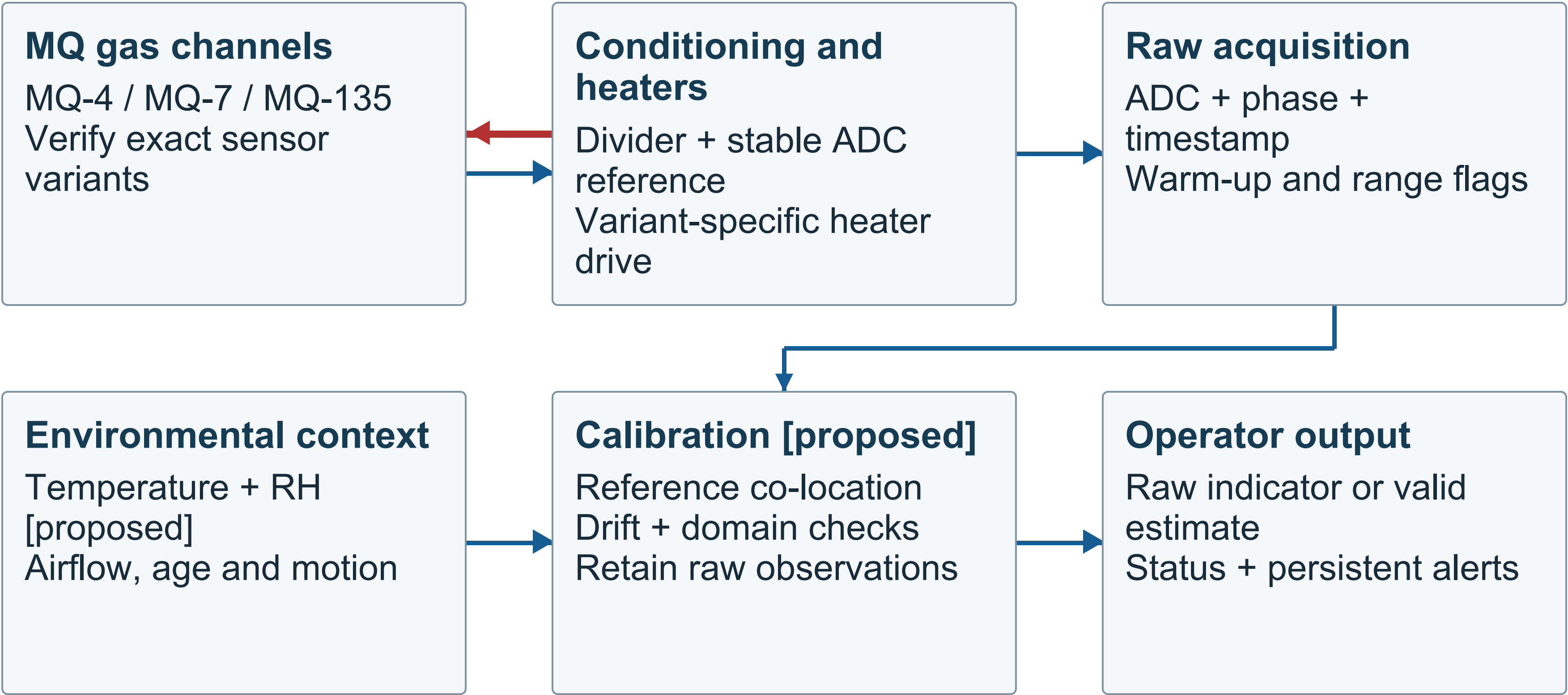
....➔ Measured state returned to control

—➔ Measurement or data record

Functional links require verification. Crossings without a junction dot are not connected.

Figure 19. Gas acquisition and calibration

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

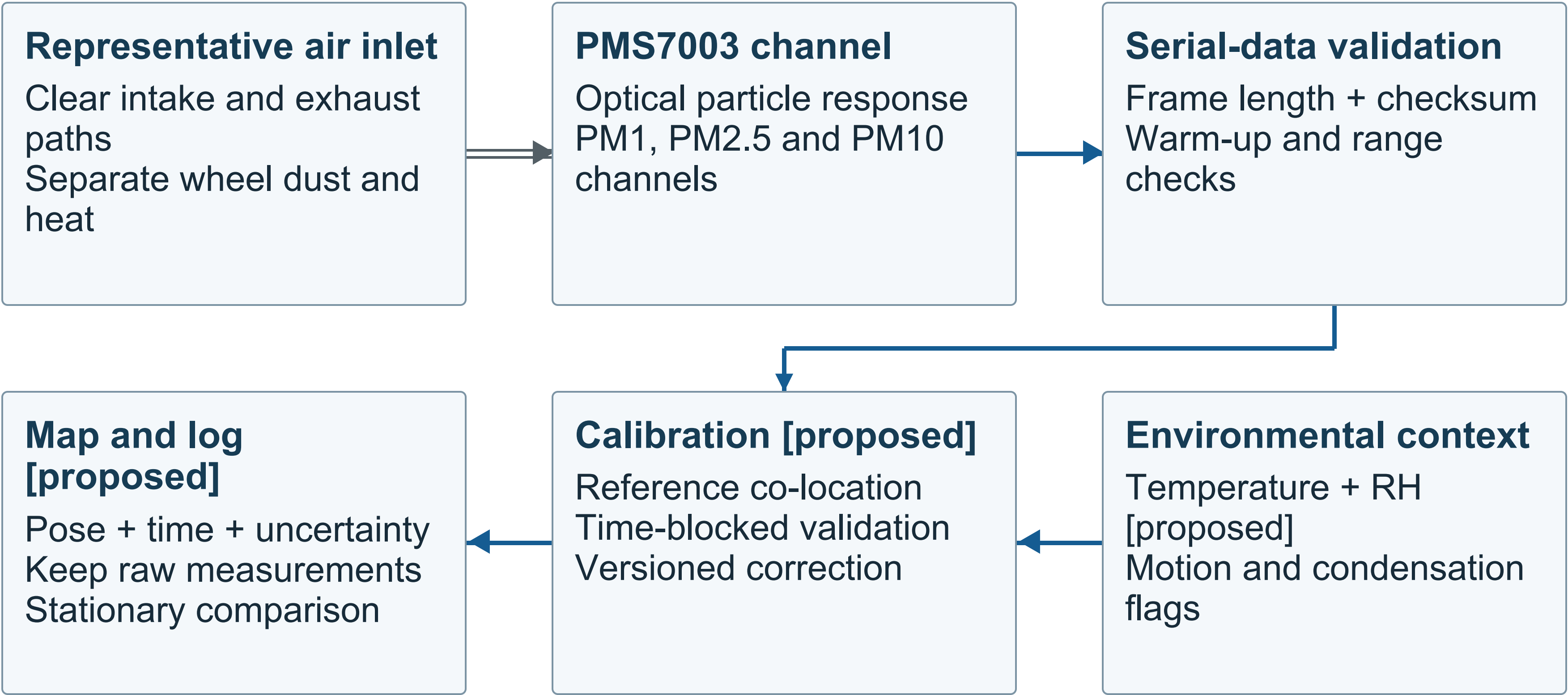
→ Measurement or data record

→ Electrical power supply

Functional links require verification. Crossings without a junction dot are not connected.

Figure 20. Particulate acquisition and calibration

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

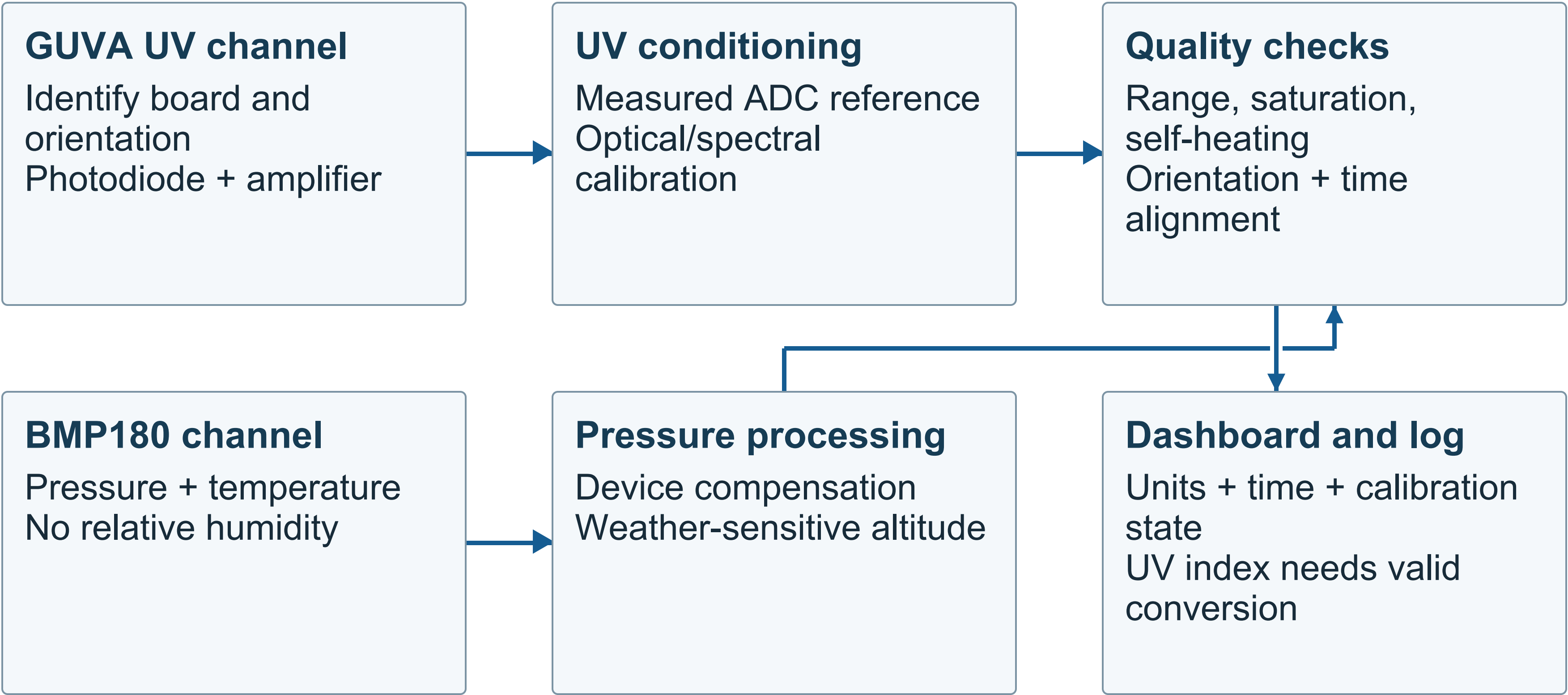
⇒ Physical transfer or interaction

→ Measurement or data record

Functional links require verification. Crossings without a junction dot are not connected.

Figure 27. UV and atmospheric measurements

A3P5 NEMESIS | Proposed functional architecture



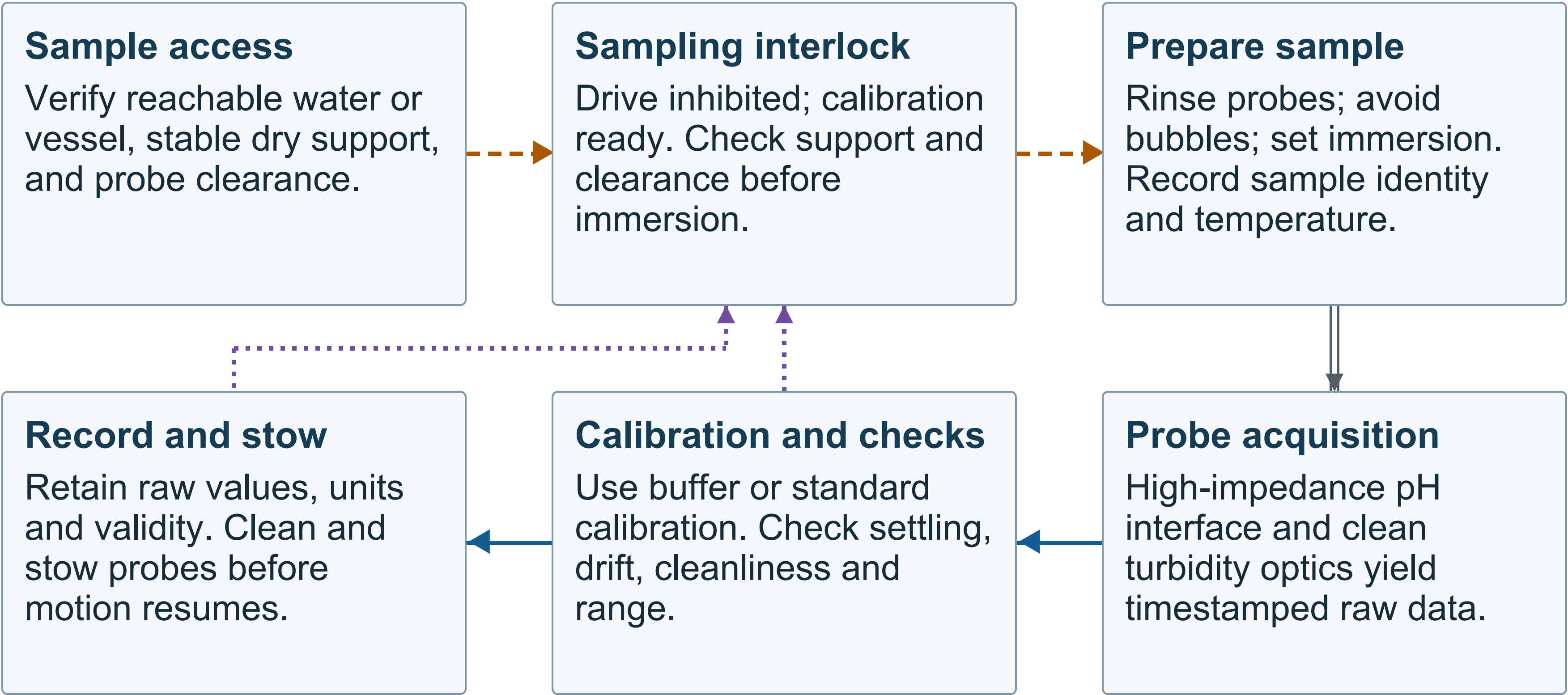
LEGEND · Arrow direction indicates the direction of the link

➡ Measurement or data record

Functional links require verification. Crossings without a junction dot are not connected.

Figure 26. Water assessment workflow

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

- - -> Permission / action

—> Measurements

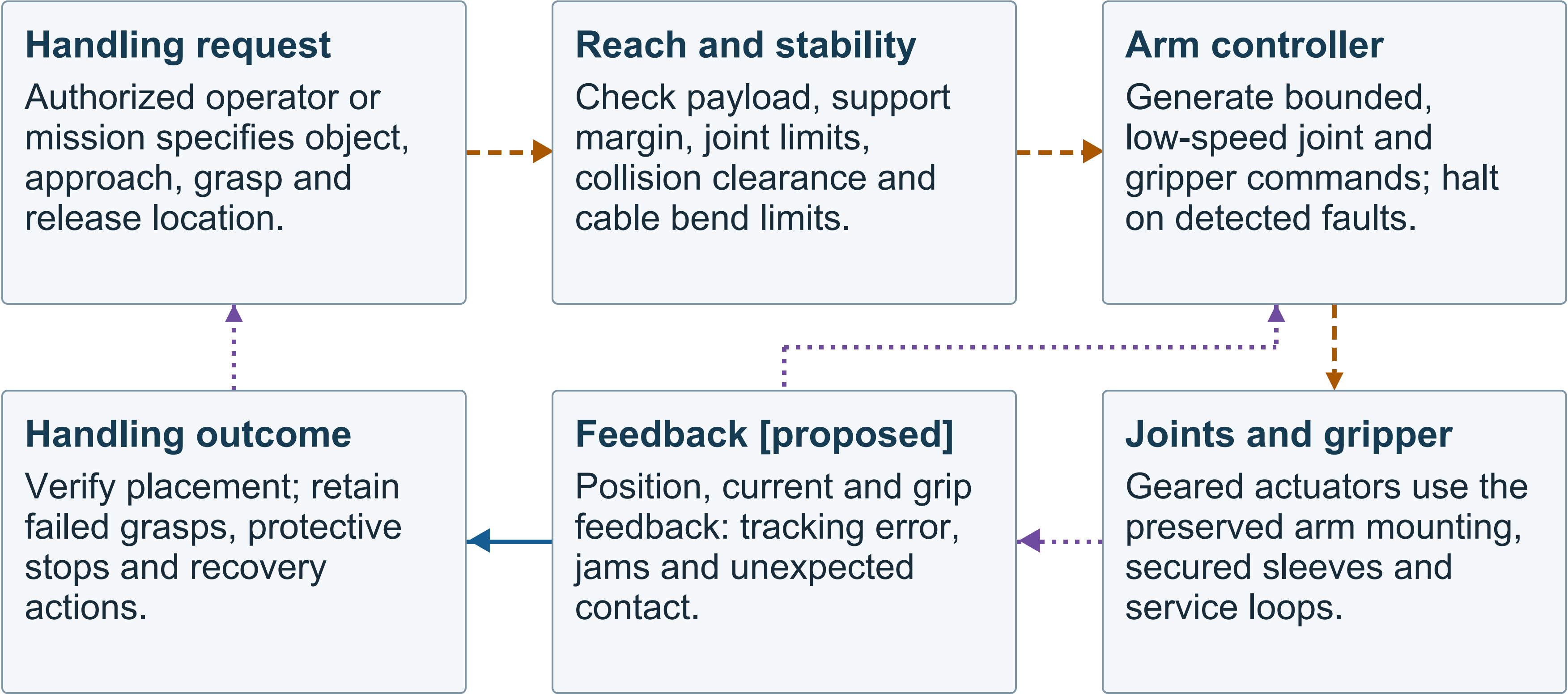
==> Sample transfer

...> Status return

Functional links require verification. Crossings without a junction dot are not connected.

Figure 13. Proposed manipulation workflow

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

Task / command

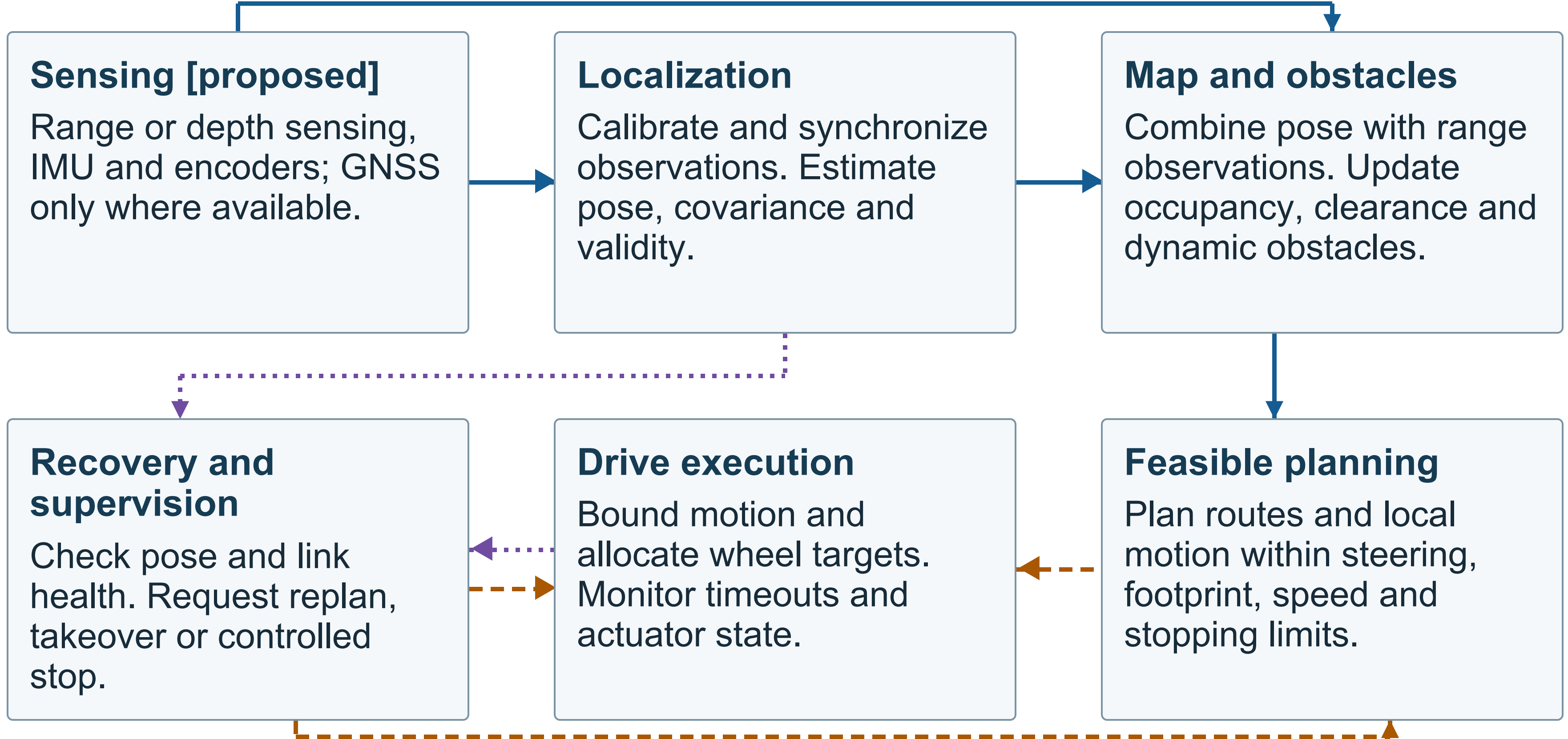
Measured response

Recorded evidence

Functional links require verification. Crossings without a junction dot are not connected.

Figure 31. Proposed navigation architecture

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

→ Observations / estimates

→ Motion / permission

State / validity

Functional links require verification. Crossings without a junction dot are not connected.

Figure 30. Communications and command authority

A3P5 NEMESIS | Proposed functional architecture

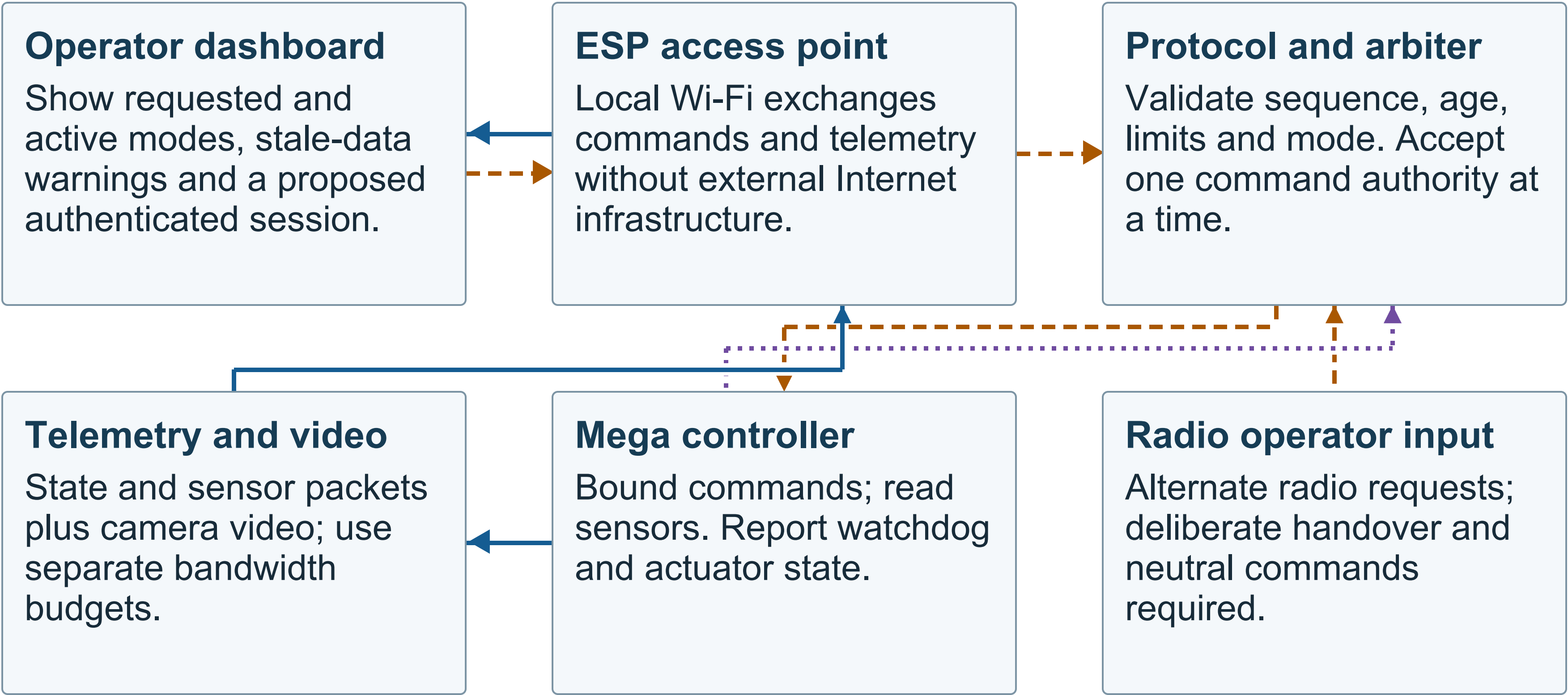
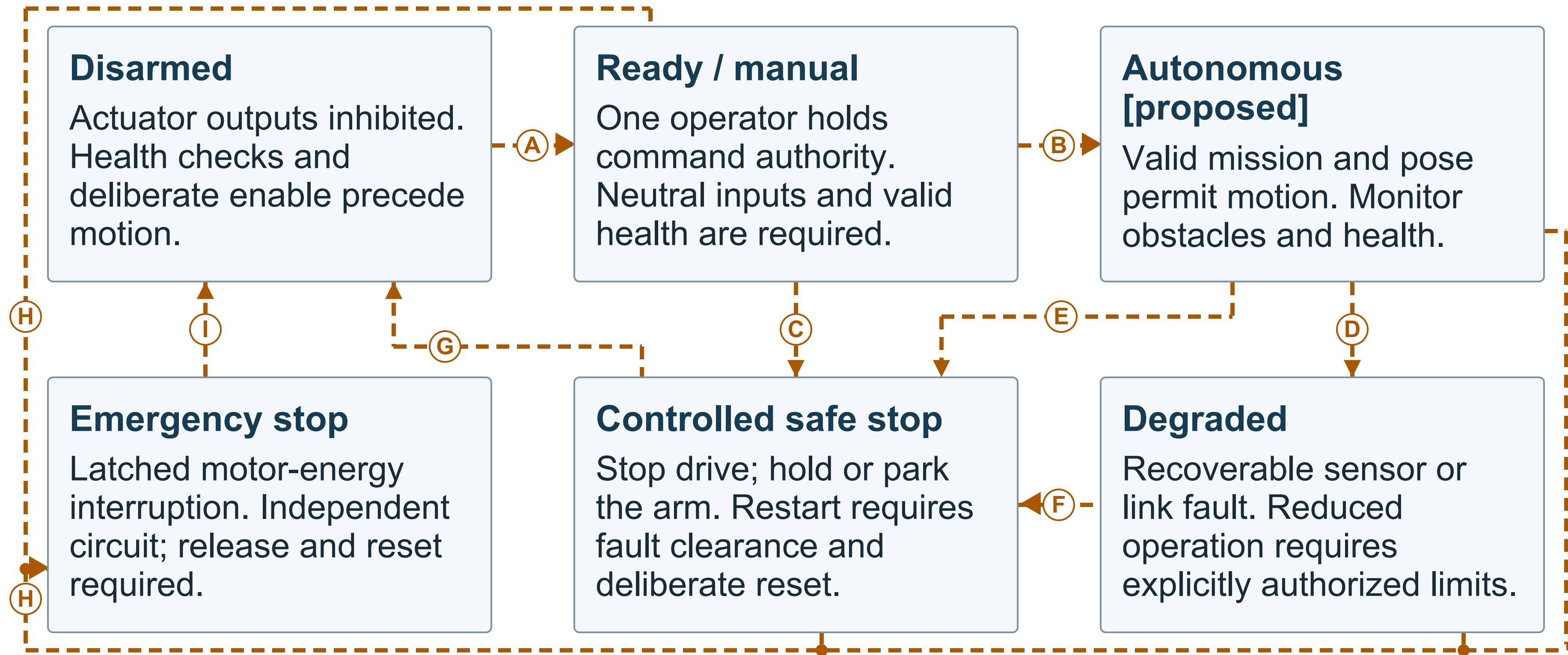


Figure 37. Operating and protective states

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Dashed arrows show guarded state transitions

A Health + enable

B Mission + pose valid

C Fault / timeout

D Recoverable fault

E Fault / takeover

F Limits unmet

G Clear fault + reset

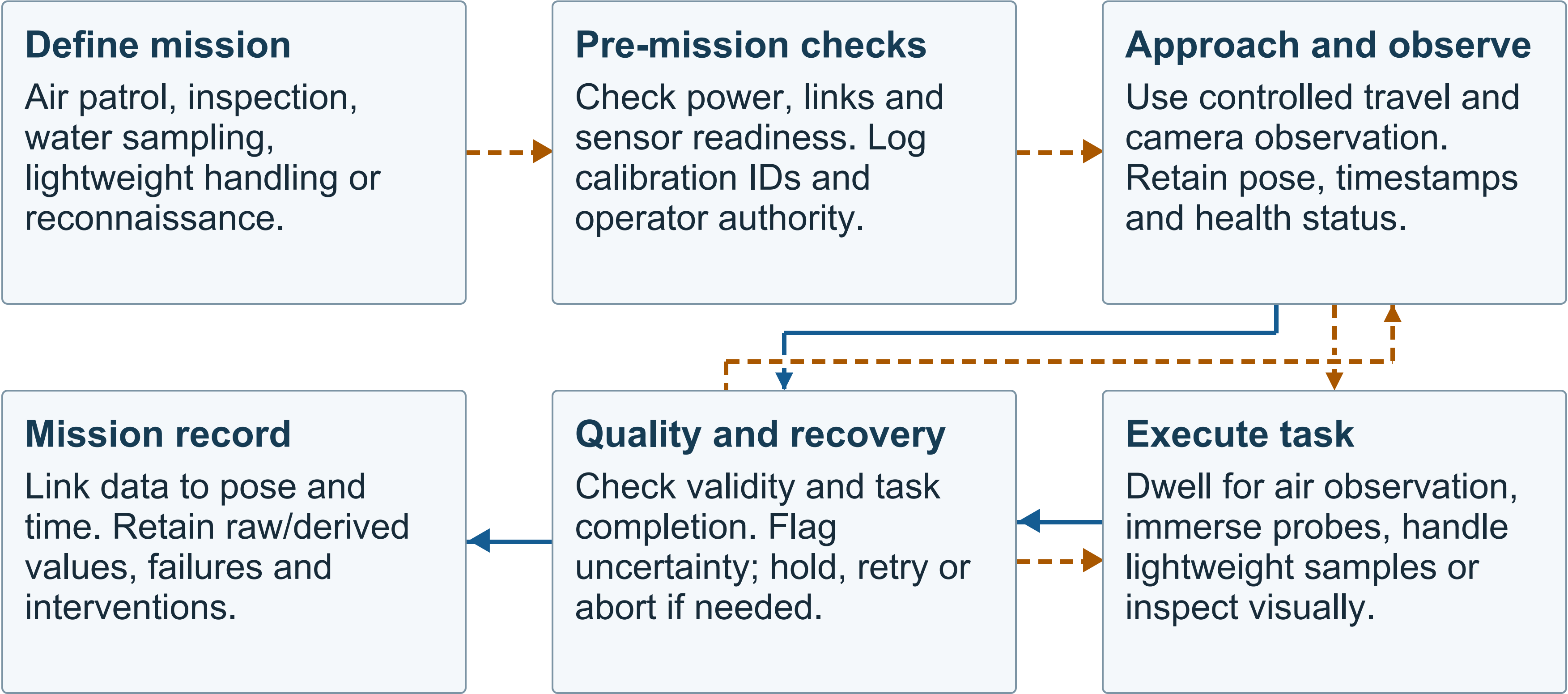
H Emergency input

I Release + reset

Dots join H emergency-input branches. Proposed transitions; emergency interruption is independent of software.

Figure 33. Mission and data workflow

A3P5 NEMESIS | Proposed functional architecture



LEGEND · Arrow direction indicates the direction of the link

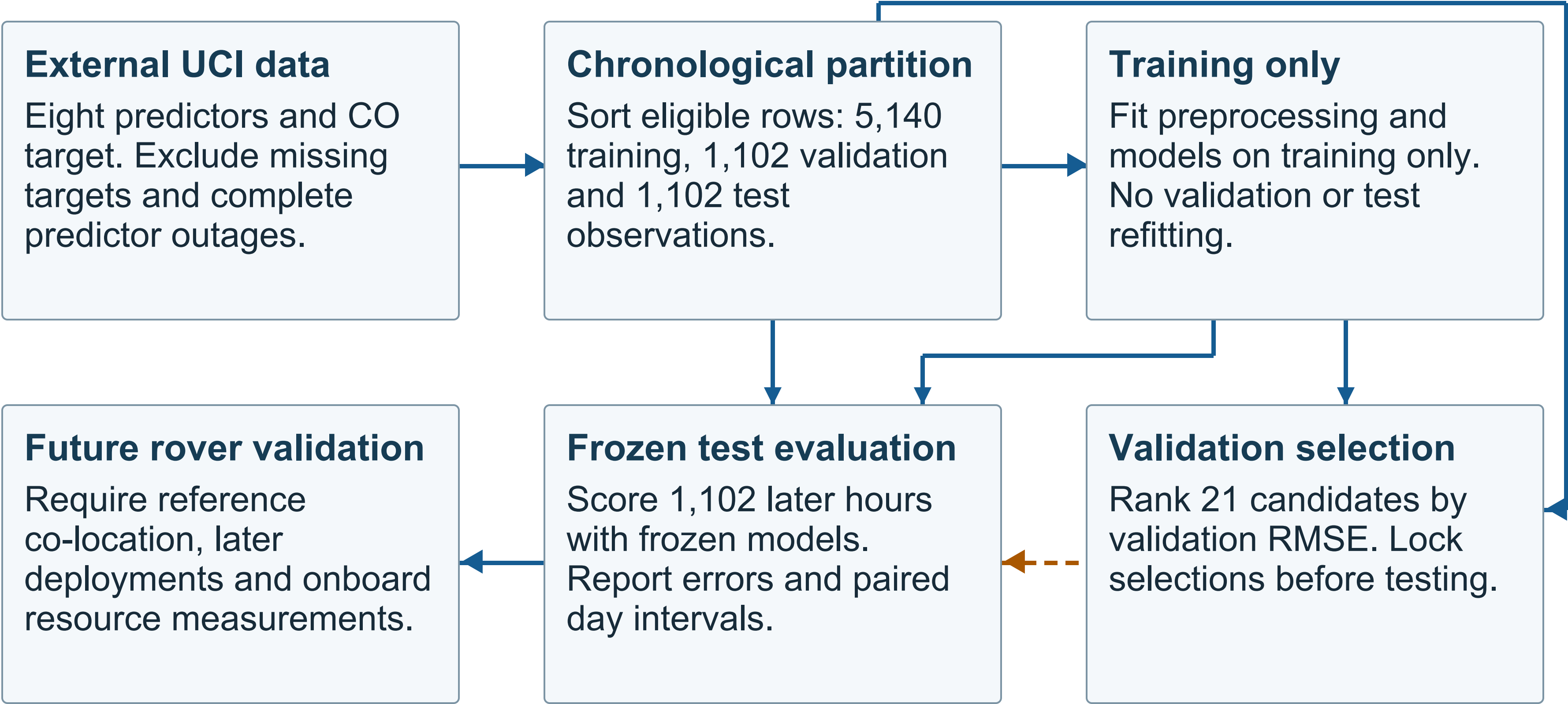
- - ➔ Task / recovery

➔ Observations / records

Functional links require verification. Crossings without a junction dot are not connected.

Figure 39. External calibration analysis pipeline

A3P5 NEMESIS | External fixed-station benchmark and future rover validation



LEGEND · Arrow direction indicates the direction of the link

→ Data / models / results - - - → Selection lock

External benchmark results do not calibrate the rover. No test-to-training feedback is permitted.